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RUA-RoutingUA

Milestone 3: Legal Requirements, Technical Risks & Business Plan

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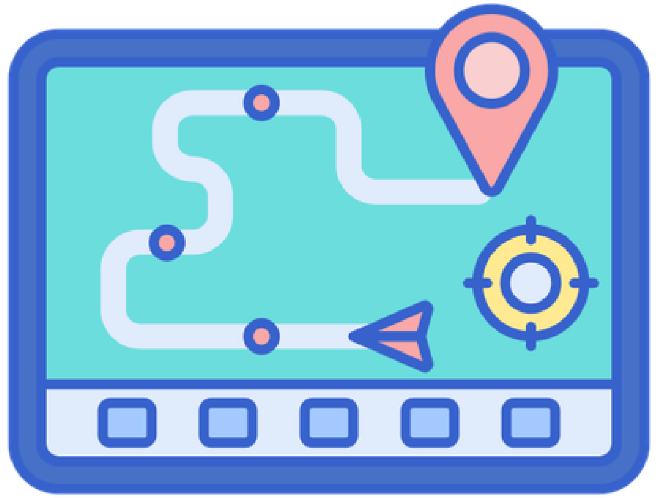
Overview

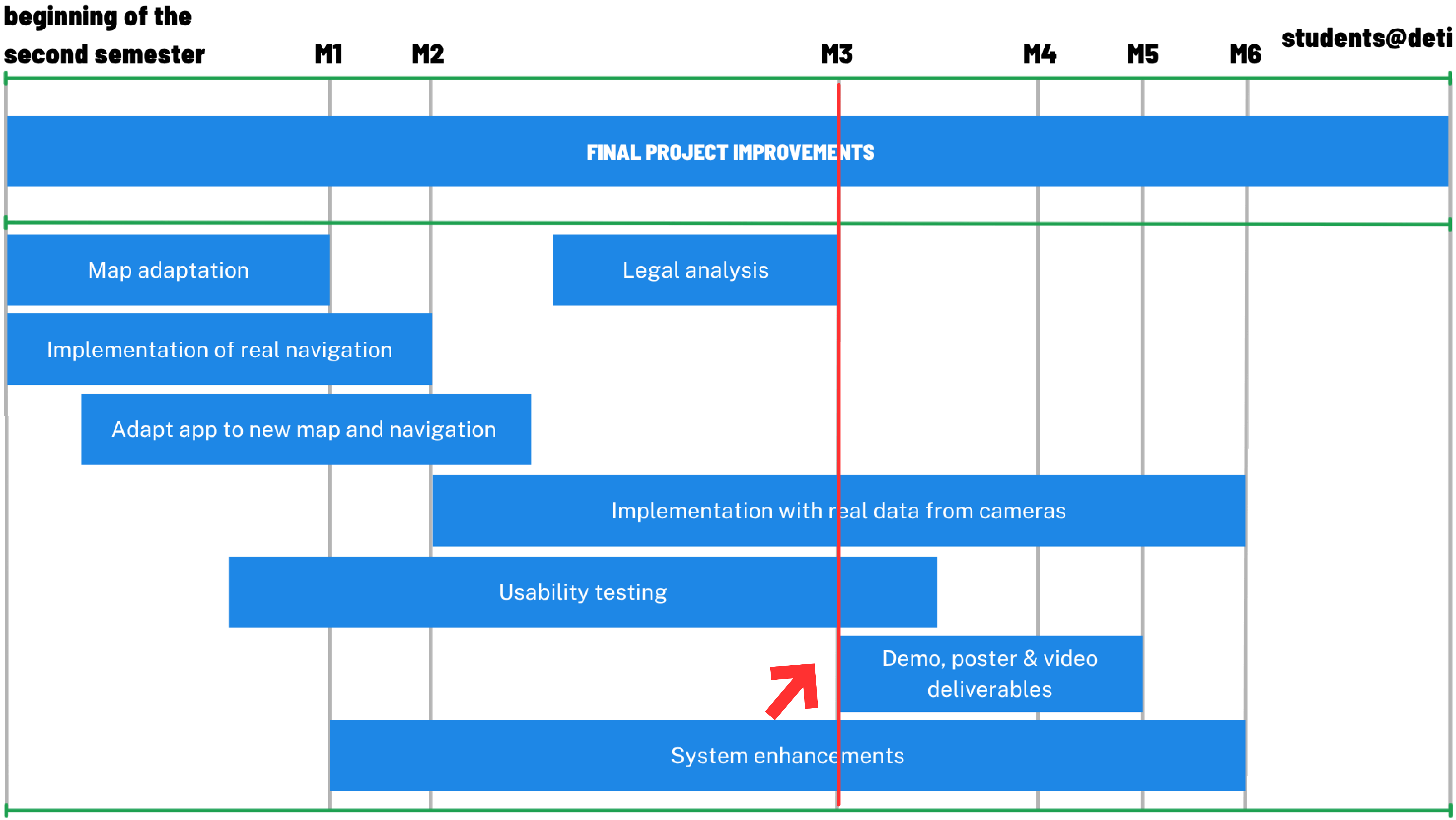
01 Context and Goals
02 Legal Requirements
03 Technical Risks
04 Technical Debt & Maintainability
05 Business Plan

Context and Goals

- Improve campus navigation for students, staff, and visitors
- Provide real-time outdoor navigation
- Ensure safety with accessible and emergency evacuation routes
- Provide real-time congestion's maps and queues of UA



Calendar



beginning of the second semester
M1
M2
M3
M4
M5
M6
students@deti
FINAL PROJECT IMPROVEMENTS
Map adaptation
Legal analysis
Implementation of real navigation
Adapt app to new map and navigation
Implementation with real data from cameras
Usability testing
Demo, poster & video deliverables
System enhancements

Legal Requirements

Data Protection (GDPR)

Collected Data

- GPS lat/lng + user_id (pseudonym)
- AI cameras collect quantitative crowd data

Purpose

- Navigation and congestion estimation
- Wait Time system

Data Protection (GDPR)

Anonymization/pseudonymization

- GPS is aggregated into 5m grid cells before publishing for congestions
- Cameras publish only counts/grid data

Data Protection (GDPR)

Data automatically deleted

- GPS users expire after 20s of inactivity
- Images data are deleted 24H after

Data Protection (GDPR)

Third-party sharing

- No sharing with third parties
- Only OSM used for public POIs (no data shared)

Data Protection (GDPR)

Security measures

Have

- Processes real-time anonymous data without storing raw footage
- Enforces a strict 5-minute expiry policy to ensure immediate data disposal

Don't have

- TLS for MQTT and encryption at rest for databases
- Formal RBAC implementation to achieve full GDPR compliance

Licensing and Intellectual Property

- OpenStreetMap - ODbL
- YOLOv8 - AGPL 3.0
- FastAPI - MIT
- PostgreSQL - PostgreSQL License
- Mosquitto - EPL 2.0/ EDL 1.0

Sector-Specific Regulation

- Camera usage permitted under public space monitoring rules(RGPD) - Camera data is only collected for quantitative purposes
- Rules about GPS data collection in public spaces(CNPD) - because the GPS data is used for quantitative purposes and routing distribution, it also expires after minutes

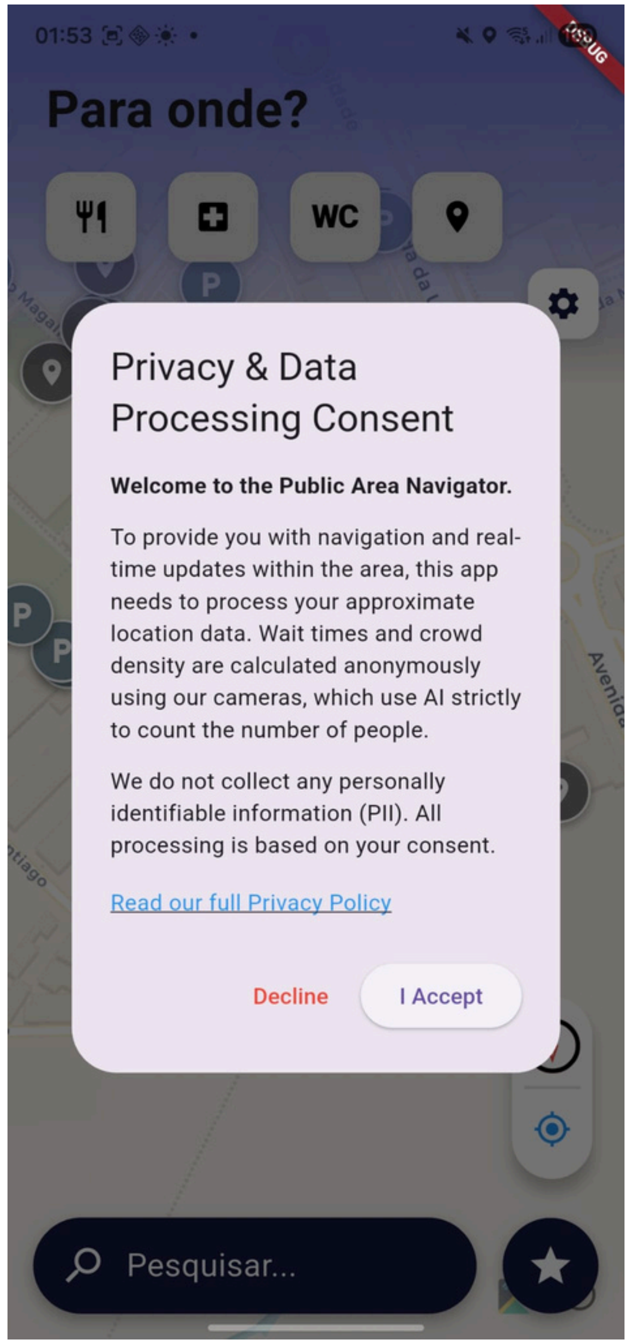
Terms and Conditions

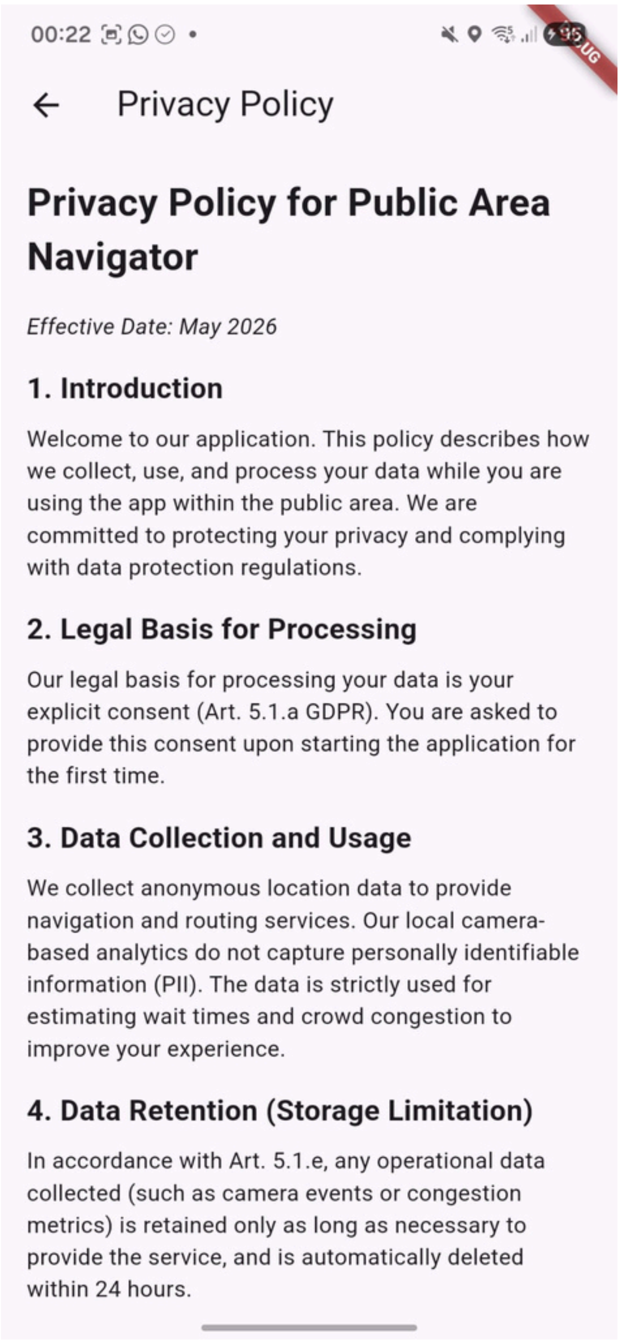
- **Data Retention:** Occupancy and queue data is retained for a maximum of 24 hours and automatically purged
- **AI Disclaimer:** Crowd estimates are generated by machine learning models and are provided "as is" — they may not reflect exact real-time conditions

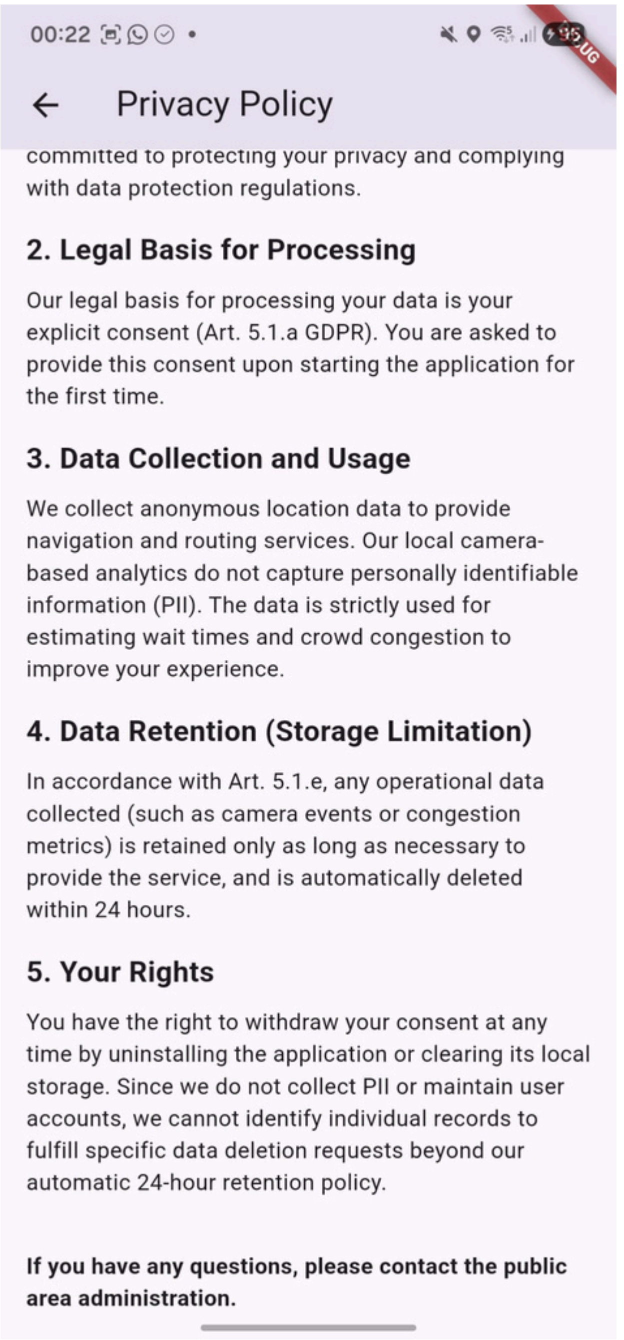
Terms and Conditions

- **Map Accuracy:** The venue/event organizer is responsible for uploading correct map data; the application team is not liable for routing errors caused by incorrect maps
- **Liability:** No responsibility for navigation mistakes, GPS signal failures, or evacuation delays caused by external technical issues

Terms and Conditions







Technical Risks

Interoperability & External Dependencies

Dependency	Risk	Mitigation
OpenStreetMap	Policy changes, rate limits, service unavailability	Local tile cache; fallback to cached map data
MQTT (Mosquitto)	Broker unavailability, bugs in paho-mqtt library	Health checks with auto-restart; WaitTime-Service already migrated to aiomqtt 2.0
PostgreSQL 15	Downtime, breaking changes in drivers	Docker health checks
Computer Vision	Model version incompatibilities (YOLOv8 - ONNX)	Triple-mode support (yolo/density/ssdlite) allows runtime fallback

Technical Debt & Maintainability

Hardcoding & Infrastructure Coupling

- **The Issue:** Backend discovery, port mapping, and MQTT topics are embedded directly in the code instead of using configuration files.

Hardcoding & Infrastructure Coupling

- **The Consequence:** The system becomes rigid and difficult to migrate, as any infrastructure change requires code modification and recompilation.

Hardcoding & Infrastructure Coupling

- **Refactoring Plan:** Centralize settings in .env files or dynamic configuration services to decouple the software from the environment.

Logic Duplication & Lack of Reusability

- **The Issue:** GPS and geolocation logic are duplicated across multiple utilities and services.

Logic Duplication & Lack of Reusability

- **The Consequence:** Increased risk of inconsistent distance calculations across services (subtle routing/POI mismatches), longer debugging cycles, and duplicated effort because the same fix must be applied in multiple places.

Logic Duplication & Lack of Reusability

- **Refactoring Plan:** Create a single, reusable geolocation module or service to ensure data consistency across the entire application.

Overcoupling

- **The Issue:** Navigation screens and frontend map components centralize too many responsibilities.

Overcoupling

- **The Consequence:** Reduces testability and significantly increases the risk of regressions where a change in one feature unexpectedly breaks another.

Overcoupling

- **Refactoring Plan:** Apply the Single Responsibility Principle by breaking large components into smaller, modular, and independent sub-components.

Fragile / Critical Components

#	Risk	Probability	Impact	Mitigation
R1	AGPL license blocks commercialization	High	Critical	Migrate to ONNX-only mode (already ⅔ modes ready)
R2	Mosquitto has no TLS/Auth	Certain	High	Enable TLS + ACLs in mosquitto.conf
R3	GPS duplication across services	Medium	Medium	Consolidate into shared geolocation module in Input-Processor
R4	paho-mqtt 1.6.1 - 2.x breaking changes	Medium	Medium	WaitTime-Service already uses aiomqtt 2.0; migrate remaining services

Business Plan

SWOT Analysis

Strengths

- A*
- M/M/1 for queueing theory
- Generic and reusable Map-Service
- Dynamic Re-routing
- Evacuation Routes
- Accessibility

S

Weaknesses

- Uneven test coverage.
- Dependency on specialized knowledge.
- Absence of explicit privacy and security governance.
- Map quality directly affects routing.

W

Opportunities

O

- Replication to other venues
- Expansion to other venue types
- Events market
- Operational analytics B2B product
- Accessibility as a market argument

Threats

T

- Tighter security and compliance requirements.
- Physical camera hardware dependency
- Operation and support costs.
- Competition from major platforms
- Drift between simulation and real environment.

TOWS Matrix

SO (Strengths + Opportunities)

B2B Licensing for European Venues

S

Multi-criteria A*, queuing theory, automated evacuation

O

High demand for real-time navigation in large-scale venues

=

A

Phased B2B rollout:

1. Local pilot
2. 5-10 venue trials
3. Full SaaS offering with dedicated support.

ST (Strengths + Threats)

MQTT Transport Hardening

S

Event-driven architecture with isolated services

T

Cybersecurity risks and lack of "enterprise readiness" for B2B deployment

=

A

Implement TLS encryption, mandatory authentication, and granular Access Control Lists

WO (Weaknesses + Opportunities)

Certified Evacuation Solutions

W

Existing emergency routes lack formal processes and official documentation.

O

Safety in large venues serves as a major product differentiator and commercial driver.

=

A

Transform evacuation features into a core commercial and compliance asset.

WT (Weaknesses + Threats)

GPS-Only "Lite" Edition

W

High hardware costs and complex camera calibration requirements

T

Market exclusion due to venue budget constraints or lack of CV infrastructure

=

A

Develop a lightweight service tier for smaller venues or pilots that functions without Computer Vision.

PESTEL Analysis

Political

Portuguese Regulatory Context

- Lei n.º 34/2013 regulates private security and video surveillance in public spaces — our camera deployment requires compliance and notifying CNPD
- Portuguese Government's Agenda Digital 2030 actively promotes Smart City infrastructure, including intelligent campus management

PESTEL Analysis

Political

EU Regulatory Landscape

- EU AI Act (Regulation 2024/1689) classifies real-time biometric surveillance as prohibited; our system avoids this by using aggregate counting only (no facial recognition)

- AI systems for crowd management in public spaces may fall under "high-risk" classification — requiring conformity assessment, transparency, and human oversight (enforcement begins 2026-2027)

PESTEL Analysis

Economic

- **Infrastructure:** near-zero cost in development
- **Deployment:** server hosting costs
- **Fixed cost:** camera hardware
- **B2B revenue potential.**

PESTEL Analysis

Social

Accessibility & Inclusion as a Core Value

- Accessibility mode to calculate routes for users with mobility challenges
- The interface includes multilingual support

Digital Literacy & Low-Friction Onboarding

- Easy to use and accessible
- Viable for older generations

PESTEL Analysis

Technological

Obsolescence Risks & Technical Debt

- Paho-mqtt 1.6.1, requires significant refactoring to upgrade to version 2.x.
- Eclipse Mosquitto 2.0 fails to utilize its TLS and authentication capabilities
- YOLOv8 necessitating a continuous update cycle

PESTEL Analysis

Environmental

Infrastructure & Carbon Footprint

- The docker configuration runs all services 24/7 as containers
- Shifting to a serverless architecture for specific services would be more energy-efficient, as resources would only be consumed during active events.

PESTEL Analysis

Legal

AGPL Licensing issues

- Technical Blocker (WT-AGPL): The Ultralytics AGPL license currently blocks proprietary CV monetization; migration to ONNX is a mandatory legal requirement

Security Certification & B2B Prerequisites

- Any B2B pilot requires TLS + Authentication to comply with Law 34/2013 (secure surveillance) and GDPR (secure transmission).

Questions?

Thank You!

Project Microsite: